

Exercise J02

Q1. Prove that: $\cos^2 \alpha + \cos^2(\alpha + \beta) - 2 \cos \alpha \cdot \cos \beta \cos(\alpha + \beta) = \sin^2 \beta$

Ans. -

Sol. -

Q2. Prove that: $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \cot \alpha$.

Ans. -

Sol. -

Q3. If $X = \sin$

$$\left(\theta + \frac{7\pi}{12}\right) + \sin\left(\theta - \frac{\pi}{12}\right) + \sin\left(\theta + \frac{3\pi}{12}\right), y = \cos\left(\theta + \frac{7\pi}{12}\right) + \cos\left(\theta - \frac{\pi}{12}\right) + \cos\left(\theta + \frac{3\pi}{12}\right)$$

then prove that $\frac{X}{Y} - \frac{Y}{X} = 2 \tan 2\theta$.

Ans. -

Sol. -

Q4. Find the positive integers p, q, r, s satisfying $\tan \frac{\pi}{24} = (\sqrt{p} - \sqrt{q})(\sqrt{r} - s)$.

Ans. $p = 3, q = 2; r = 2; s = 1$

Sol. -

Q5. If the value of the expression $\sin 25^\circ \cdot \sin 35^\circ \cdot \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$ where a, b, c $\in \mathbb{N}$ and are in their lowest form, find the value of (a + b + c).

Ans. 24

Sol. -

Q6. Prove that: (a) $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ = 3$

(b) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ = 4$. (c)

$$\sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16} = \frac{3}{2}$$

Ans. -

Sol. -

Q7. If α, β and γ are in A.P., show that $\cot \beta = \frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha}$

Ans. -

Sol. -

Q8. If $\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$, prove that $\sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

Ans. -

Sol. -

Q9. If $m \tan (\theta - 30^\circ) = n \tan (\theta + 120^\circ)$, show that $\cos 2\theta = \frac{m+n}{2(m-n)}$..

Ans. -

Sol. -

Q10. Let $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$
then find the value of (a) $P(5)$ and (b) $P(6)$.

Ans. (a) $\frac{3-\sqrt{5}}{32}$; (b) $\frac{2-\sqrt{3}}{16}$

Sol. -

Q11. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ).....(1 + \tan 45^\circ) = 2^n$, find n.

Ans. $n = 23$

Sol. -

Q12. The value of : $\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$ is equal to

Ans. $\frac{1}{64}$

Sol. -

Q13. $\sin (B + 2C) + \sin (C + 2A) + \sin (A + 2B) = 4 \sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2}$

Ans. -

Sol.
$$\begin{aligned} &= 4 \sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2} \sin(B+2C) + \sin(C+2A) + \sin(A+2B) \\ &= \sin(B+C+C) + \sin(C+A+A) + \sin(A+B+B) \\ &= \sin(\pi - A + C) \sin(\pi - B + A) + \sin(\pi - C + B) \\ &= \sin(A-C) + \sin(B-A) + \sin(C-B) \\ &= 2 \sin \frac{B-C}{2} \cos \frac{2A-B-C}{2} + 2 \sin \frac{C-B}{2} \cos \frac{C-B}{2} \\ &= 2 \sin \frac{B-C}{2} \left\{ \cos \frac{2A-B-C}{2} - \cos \frac{B-C}{2} \right\} \\ &= 4 \sin \frac{B-C}{2} \sin \frac{A-C}{2} \sin \frac{B-A}{2} \end{aligned}$$

Q14. $\cos \frac{A}{2} + \cos \frac{B}{2} - \cos \frac{C}{2} = 4 \cos \frac{\pi + A}{4} \cos \frac{\pi + B}{4} \cos \frac{\pi - C}{4}$

Ans. -

Sol. $4 \cos \frac{\pi + A}{4} \cos \frac{\pi + B}{4} \cos \frac{\pi - C}{4} = 2 \cos \frac{\pi - C}{4} \left\{ \cos \frac{2\pi + A + B}{4} + \cos \frac{A - B}{4} \right\}$
 $2 \cos \frac{A + B}{4} \left\{ \cos \left(\frac{\pi}{2} + \frac{A + B}{4} \right) + \cos \left(\frac{A - B}{4} \right) \right\}$
 $- 2 \cos \frac{A + B}{4} \sin \frac{A + B}{4} + 2 \cos \frac{A + B}{4} \cos \frac{A - B}{4}$
 $-\sin \frac{A + B}{2} + \cos \frac{A}{2} + \cos \frac{B}{2}$
 $\cos \frac{A}{2} + \cos \frac{B}{2} - \cos \frac{C}{2}$

Q15. Show that $2^{\sin x} + 2^{\cos x} \geq 2^{1 - \frac{1}{\sqrt{2}}}$

Ans. -

Sol. -

Q16. Show that $\frac{3 + \cos x}{\sin x} \forall x \in R$ can not have any value between $-2\sqrt{2}$ and $2\sqrt{2}$. What inference can you draw about the values of ?

Ans. $\left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right]$

Sol. -

Q17. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is

Ans. 2

Sol. $\frac{1}{4 \cos^2 \theta + 1 + \frac{3}{2} \sin 2\theta}$
 $\Rightarrow \frac{1}{2[1 + \cos 2\theta] + 1 + \frac{3}{2} \sin 2\theta}$

lies between $\frac{1}{2}$ to $\frac{11}{2}$

$\therefore$ maximum value is 2.

Minimum value of $1 + 4 \cos^2 \theta + 3 \sin \theta \cos \theta$

$1 + \frac{4(1 + \cos 2\theta)}{2} + \frac{3}{2} \sin 2\theta$
 $= 1 + 2 + 2 \cos 2\theta + \frac{3}{2} \sin 2\theta$

$$= 3 + 2 \cos 2\theta + \frac{3}{2} \sin 2\theta$$

$$\therefore = 3 - \sqrt{4 + \frac{9}{4}} = 3 - \frac{5}{2} = \frac{1}{2}$$

So maximum value of $\frac{4}{4 \cos^2 \theta + 1 + \frac{3}{2} \sin 2\theta}$ is 2.

Q18. If $f(\theta) = \sum_{n=1}^6 \operatorname{cosec}\left(\theta + \frac{(n-1)}{4}\right) \operatorname{cosec}\left(\theta + \frac{n\pi}{4}\right)$, where $0 < \theta < \frac{\pi}{2}$, then find the minimum value of $f(\theta)$.

Ans. $2\sqrt{2}$

Sol. -

Q19. Let $k = 1^\circ$, then prove that $\sum_{n=0}^{88} \frac{1}{\cos nk \cdot \cos(n+1)k} = \frac{\cos k}{\sin^2 k}$

Ans. -

Sol. -

Q20. Prove that : $\operatorname{cosec} \theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 2^2 \theta + \dots + \operatorname{cosec} 2^{n-1} \theta = \cot(\theta/2) - \cot 2^{n-1} \theta$.

Ans. -

Sol. -

Q21. Let $f_n(\theta) = \sum_{n=1}^n \tan\left(\frac{\theta}{2^n}\right) \sec\left(\frac{\theta}{2^{n-1}}\right)$, then compute the value of
(a) $f_3(2\pi)$, (b) $f_4\left(\frac{4\pi}{3}\right)$ (c) $f_5(4\pi)$ and (d) $f_6(48\pi)$.

Ans. (a) -1 , (b) $2(\sqrt{3} - 1)$, (c) $1 - \sqrt{2}$, (d) 1

Sol. -

Q22. The minimum value of $16 \cos^4 \theta - 20 \cos^2 \theta + 5$ is :

(A) 1 (B) $-\frac{5}{4}$ (C) 0 (D) None of these

Ans. B

Sol. -

Q23. The maximum and minimum values of $|\sin x| + \cos x$ when $x \in \mathbb{R}$ must be :

(A) $[0, 1]$ (B) $[-\sqrt{2}, \sqrt{2}]$ (C) $[-\sqrt{2}, 1]$ (D) $[0, \sqrt{2}]$

Ans. B

Sol. -

Q24. The range of the function $|\sin x| + |\cos x|$ must be :

- (A) $[-\sqrt{2}, \sqrt{2}]$ (B) $[1, \sqrt{2}]$ (C) $[0, \sqrt{2}]$ (D) None of these

Ans. B

Sol. -

Q25. If $\sec \theta + \tan \theta = 1$, then one root of equation, $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is :

- (A) $\tan \theta$ (B) $\sec \theta$ (C) $\cos \theta$ (D) $\sin \theta$

Ans. C, B

Sol. -

Q26. If $\sin \alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$ and $\cos \beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$, then the correct statement(s) is(are) :

- (A) $\tan(\alpha - \beta) = \frac{63}{16}$ (B) $\tan(\alpha + \beta) = \frac{33}{56}$ (C) $\sin 2\alpha = \frac{-24}{25}$ (D) $\cos 2\beta = \frac{-119}{169}$

Ans. A, B, C, D

Sol. $\sin \alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$

$$\Rightarrow \cos \alpha = -\sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{-4}{5}$$

$$\text{Also, } \cos \beta = \frac{-5}{13}, \pi < \beta < \frac{3\pi}{2}$$

$$\Rightarrow \sin \beta = -\sqrt{1 - \cos^2 \beta} = -\sqrt{1 - \frac{25}{169}} = \frac{-12}{13}$$

Now,

$$(A) \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{\left(\frac{-3}{4}\right) - \left(\frac{12}{5}\right)}{1 + \left(\frac{-3}{4}\right)\left(\frac{12}{5}\right)} = \frac{\frac{-15-48}{20}}{\frac{20-36}{20}} = \frac{-63}{-16} = \frac{63}{16}$$

$$(B) \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\frac{-3}{4} + \left(\frac{12}{5}\right)}{1 - \left(\frac{-3}{4}\right)\left(\frac{12}{5}\right)} = \frac{\frac{-15+48}{20}}{\frac{20+36}{20}} = \frac{33}{56}$$

$$(C) \sin 2\alpha = 2 \sin \alpha \cos \alpha = 2 \left(\frac{3}{5}\right) \left(\frac{-4}{5}\right) = \frac{-24}{25}$$

$$(D) \cos 2\beta = 2\cos^2 \beta - 1 = 2 \times \frac{25}{169} - 1 = \frac{50 - 169}{169} = \frac{-119}{169}. \text{ Ans.}]$$

Q27. If $\sin x = \frac{1}{4}(\sqrt{5} - 1)$ then which of the following is/are correct?

- (A) $\cos 4x = \sin x$ (B) $\sin 3x = \cos 2x$ (C) $\sin 2x + \cos 3x = \frac{1}{2}$

$$(D) \sin 5x + \cos 5x = 1$$

Ans. A, B, D

Sol. $\sin x = \frac{\sqrt{5} - 1}{4}$

$$\Rightarrow x = 18^\circ$$

$$(A) \cos 4x = \cos 72^\circ = \sin 18^\circ$$

$$(B) \cos 2x = \cos 36^\circ = \sin (54^\circ) = \sin 3x$$

$$(C) \sin 36^\circ + \cos 54^\circ = \sin 36^\circ + \sin 36^\circ = 2\sin 36^\circ = 4 \times \sin 18^\circ \cos 18^\circ$$

$$= 4 \times \frac{\sqrt{5} - 1}{4} \times \frac{\sqrt{5} + 1}{4} = 1$$

$$(D) \sin 90^\circ + \cos 90^\circ = 1$$

Q28. If the equation $2 \cos^2 x + \cos x - a = 0$ has solutions then a can be :

(A) $\frac{-1}{4}$ (B) $\frac{-1}{8}$ (C) 2 (D) 5

Ans. B, C

Sol. $2 \cos^2 x + \cos x - a = 0$

$$\Rightarrow \cos x = \frac{-1 \pm \sqrt{1 + 8a}}{4}$$

$$\Rightarrow \text{For real root } 1 + 8a \geq 0$$

$$\Rightarrow a \geq \frac{-1}{8}$$

$$(C) \text{ If } a = 2$$

$$\text{then } \cos x = \frac{-1 \pm \sqrt{17}}{4}$$

$$\text{then } \cos x = \frac{-1 \pm \sqrt{41}}{4} \in (-\infty, -1) \cup (1, +\infty)$$

Q29. $\frac{\sin^2 50^\circ - \sin^2 40^\circ}{\cos^2 50^\circ \cdot \cos^2 40^\circ}$ is equal to :

(A) $4 \tan 10^\circ \sec 10^\circ$ (B) $2 \tan 10^\circ \sec 10^\circ$ (C) $4 \tan 10^\circ (\tan 10^\circ + \cot 40^\circ)$

(D) $4 \tan 10^\circ (\tan 10^\circ + \tan 40^\circ)$

Ans. A, D

Sol.
$$\frac{\sin^2 50^\circ - \sin^2 40^\circ}{\cos^2 50^\circ \cdot \cos^2 40^\circ} = \frac{4(\sin^2 50^\circ - \sin^2 40^\circ)}{(2 \cos 50^\circ \cdot \cos 40^\circ)^2} = \frac{4 \sin 90^\circ \cdot \sin 10^\circ}{(\cos 90^\circ + \cos 10^\circ)^2}$$

$$= \frac{4 \sin 10^\circ}{\cos 10^\circ \cdot \cos 10^\circ}$$

$$= 4 \tan 10^\circ \sec 10^\circ$$

$$\begin{aligned} \text{now (D)} &\rightarrow 4 \tan 10^\circ \left[\frac{\sin 10^\circ \cos 40^\circ + \sin 40^\circ \cos 10^\circ}{\cos 10^\circ \cdot \cos 40^\circ} \right] \\ &= 4 \tan 10^\circ \left[\frac{\sin 50^\circ}{\cos 10^\circ \cdot \cos 40^\circ} \right] \\ &= 4 \tan 10^\circ \frac{1}{\cos 10^\circ} = 4 \tan 10^\circ \sec 10^\circ \end{aligned}$$

Q30. Which of following functions have the maximum value unity ?

(A) $\sin^2 x - \cos^2 x$ (B) $\sqrt{\frac{1}{\sqrt{2}}} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{3}} \cos x \right)$ (C) $\cos^6 x + \sin^6 x$
 (D) $\cos^2 x + \sin^4 x$

Ans. D

Sol. -

Q31. If $\cos \beta$ is geometric mean between $\sin \alpha$ and $\cos \alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\cos 2\beta$ =

(A) $-2 \sin^2 \left(\frac{\pi}{2} - \alpha \right)$ (B) $-2 \cos^2 \left(\frac{\pi}{4} + \alpha \right)$ (C) $2 \sin^2 \left(\frac{\pi}{4} + \alpha \right)$ (D) $2 \cos^2 \left(\frac{\pi}{4} - \alpha \right)$

Ans. B

Sol. -

Q32. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then :

(A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$ (C) $\tan^2 x = \frac{1}{3}$
 (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

Ans. A, B

Sol. $\frac{\tan^4 x}{2} + \frac{1}{3} = \frac{\sec^4 x}{5}$

Let $\tan^2 x = t$

$$\Rightarrow \frac{t^2}{2} + \frac{1}{3} = \frac{(1+t)^2}{5}$$

$$\Rightarrow \frac{3t^2 + 2}{6} = \frac{t^2 + 2t + 1}{5}$$

$$\Rightarrow t = \frac{2}{3}$$

$$\Rightarrow \sin^2 x = \frac{2}{5} \text{ \& } \cos^2 x = \frac{3}{5}$$

$$\Rightarrow \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$$

Q33. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of

$$\sum_{m=1}^6 \operatorname{cosec} \left(\theta + \frac{(m-1)\pi}{4} \right) \operatorname{cosec} \left(\theta + \frac{m\pi}{4} \right) = 4\sqrt{2} \text{ is (are) :}$$

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

Ans. C, D

Sol.
$$\sqrt{2} \sum_{m=1}^6 \frac{\sin \left[\left(\theta + \frac{m\pi}{4} \right) - \left(\theta + \frac{m\pi}{4} - \frac{\pi}{4} \right) \right]}{\sin \left(\theta + \frac{m\pi}{4} - \frac{\pi}{4} \right) \sin \left(\theta + \frac{m\pi}{4} \right)} = 4\sqrt{2}$$

$$\Rightarrow \sqrt{2} \sum_{m=1}^6 \left[\cot \left(\theta + \frac{(m-1)\pi}{4} \right) - \cot \left(\theta + \frac{m\pi}{4} \right) \right] = 4\sqrt{2}$$

$$\Rightarrow \cot \theta + \tan \theta = 4.$$

$$\Rightarrow \tan \theta = 2 \pm \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{12} \text{ or } \frac{5\pi}{12}$$

Q34. If α and β satisfy the equation, $a \cos \theta + b \sin \theta = c$, then

Column I

(A) the value of $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2}$ is

(B) the value of $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}$ is

(C) the value of $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2}$ is

(D) the value of $\left(\tan \frac{\alpha}{2} - \tan \frac{\beta}{2} \right)^2$ is

Column II

(P) $\frac{2b}{c+a}$

(Q) $\frac{c-a}{c+a}$

(R) $\frac{4(b^2 + a^2 - c^2)}{(c+a)^2}$

(S) $\frac{2(2b^2 + a^2 - c^2)}{(c+a)^2}$

(A) (A) Q (B) P (C) S (D) R (B) (A) R (B) Q (C) S (D) R (C) (A) Q (B) P (C) R (D) S

(D) (A) Q (B) S (C) P (D) R

Ans. 5ef5c36d9fff15766e2fb3be

Sol. -

Q35.

Column-I

Column-II

(A) The value of $\operatorname{cosec} \frac{\pi}{18} - \sqrt{3} \sec \frac{\pi}{18}$ is a

(P) 1

(B) $\frac{\cos 20^\circ + 8 \sin 70^\circ \sin 50^\circ \sin 1^\circ}{\sin^2 80^\circ}$ is equal to

(Q) 2

(C) Exact value of $\cos 20^\circ + 2 \sin 55^\circ - \sqrt{2} \sin 65^\circ$ is

(R) 3

(D) Let $f(\theta) = \frac{1}{2} + \frac{2}{3} \operatorname{cosec}^2 \theta + \frac{3}{8} \sec^2 \theta$.

(S) 4

The least value of $f(\theta)$ for all permissible values of θ , is greater than

(A) (A) S (B) Q (C) PS (D) Q (B) (A) S (B) Q (C) P (D) PQ

(C) (A) Q (B) S (C) P (D) Q (D) (A) S (B) P (C) Q (D) QS

Ans. 5ef5c5a1f6c4ee7669672293

Sol.

$$(A) \frac{1}{\sin \pi/18} = \frac{\sqrt{3}}{\cos \pi/18} = \frac{2 \left[\frac{1}{2} \cos \frac{\pi}{18} - \frac{\sqrt{3}}{2} \sin \frac{\pi}{18} \right]}{\sin(\pi/9)}$$

$$= \frac{4 \left[\sin \frac{\pi}{6} - \cos \frac{\pi}{6} \sin \frac{\pi}{18} \right]}{\sin \frac{\pi}{9}} = 4 \text{ Ans.}$$

$$(B) \text{Nr.} = \cos 20^\circ + 4(\cos 20^\circ - \cos 120^\circ) \sin 10^\circ$$

$$= \cos 20^\circ + 2(\sin 30^\circ - \sin 10^\circ) - 4(-1/2) \sin 10^\circ$$

$$= 1 + \cos 20^\circ = 2 \cos^2 10^\circ \Rightarrow 2 \cos^2 10^\circ / \sin^2 80^\circ = 2$$

$$(C) \cos 20^\circ + 1 - \cos 110^\circ - \sqrt{2} \sin 65^\circ$$

$$2 \sin 65^\circ \sin 45^\circ - \sqrt{2} \sin 65^\circ + 1 = 1 \text{ Ans.}$$

$$(D) \frac{1}{2} + \frac{2}{3} + \frac{3}{8} + \frac{2}{3} \cot^2 \theta + \frac{3}{8} \tan^2 \theta \geq \frac{37}{24} + 1 = \frac{61}{24} \text{ by AM-GM in equality.}$$